Arnab Roy

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Google Scholar: scholar.google.com/citations?user=NcPpNqQAAAAJ | GitHub: github.com/arn97

Research Interests

- Scientific Machine Learning
- Diffusion Models and Simulation-Based Inference
- Operator Learning and Neural Operators
- Inverse Problems and Parameter Estimation for PDEs and ODEs
- Computational Modeling of Dynamical Systems
- Numerical Analysis of Differential Equations

Education

University of Delaware, Newark, DE, USA *August 2022 – Expected June 2027*
PhD in Applied Mathematics + Master's in Data Science, Department of Mathematical Sciences
Advisor: Dr. Tobin A. Driscoll
GPA: 3.746/4.000

Indian Institute of Science Education and Research (IISER) Berhampur, India *August 2016 – May 2021*
BS-MS Dual Degree (Major: Mathematics, Minor: Physics)
CPI: 9.44/10.00
Advisor: Dr. Ritabrata Sengupta
Thesis: *Interpolation by Different Types of Quantum Channels Using Conic Programs and Further Generalisation*

Publications and Preprints

Journal Articles

Chen, Q., **Roy, A.**, and Driscoll, T. A. *Operator learning for models of tear film breakup. Mathematical Medicine and Biology: A Journal of the IMA*, 2026. Advance access, article dqag009.
DOI: https://doi.org/10.1093/imammb/dqag009 | arXiv: 2601.08001

Preprints

Roy, A., and Driscoll, T. A. *Amortized Generative Modeling of Invariant Measures across A Family of Kuramoto–Sivashinsky PDEs.* arXiv: 2609.32914, 2026. Under review, ICLR 2027.
DOI: https://doi.org/10.48550/arXiv.2609.32914

Roy, A. and Patra, S. *Interpolation by Different Types of Quantum Channels Using Conic Programs and Further Generalisation.* arXiv: 2104.07254, 2021.
DOI: https://doi.org/10.48550/arXiv.2104.07254

Research Experience

Stochastic Operator Learning for Tear Film Dynamics Using Diffusion Models *2025 – Present*
University of Delaware, PhD Research under Dr. Tobin A. Driscoll

- Developing diffusion-based simulation-based inference models to learn posterior distributions of parameters governing tear film ODE systems from observed fluorescence intensity data.
- Implementing conditional diffusion models using BayesFlow and PyTorch for probabilistic parameter inference.

GitHub: Tearfilm_SBI_diffusion_ODE

Amortized Diffusion Model for Kuramoto–Sivashinsky PDE *2025 – 2026*
University of Delaware, PhD Research

- Developed a diffusion model that generates samples from the invariant measure of the Kuramoto–Sivashinsky equation, amortized across a family of PDE parameters so that a single trained network spans regimes from steady to fully chaotic.
- Posted as an arXiv preprint and submitted to ICLR 2027; currently under review.

Paper: doi.org/10.48550/arXiv.2609.32914 | GitHub: Diffusion_forecast_KS

Operator Learning for Models of Tear Film Breakup *2023 – 2025*
University of Delaware, PhD Research under Dr. Tobin A. Driscoll

- Developed neural operator models to learn mappings from fluorescence intensity to tear film thickness and osmolarity.
- Implemented Fully-connected Neural Nets, Fourier Neural Nets, PCA-Neural Nets and related architectures as surrogate models for dynamical systems governed by differential equations.
- Published in Mathematical Medicine and Biology: A Journal of the IMA (2026).

Paper: doi.org/10.1093/imammb/dqag009 | GitHub: sciml-tearfilms

Interpolation by Quantum Channels Using Conic Programming *2020 – 2021*
IISER Berhampur, Undergraduate Thesis under Dr. Ritabrata Sengupta

- Developed optimization-based methods to determine existence and construction of quantum channels interpolating prescribed input-output quantum states.
- Applied conic programming techniques to characterize feasible interpolation operators.

Paper: doi.org/10.48550/arXiv.2104.07254

Teaching Experience

Instructor of Record — Department of Mathematical Sciences, University of Delaware

MATH 351: Engineering Mathematics I *Fall 2026 (34 students)*
MATH 351: Engineering Mathematics I *Summer 2026 (8 students)*
MATH 305: Applied Mathematics for Biomedical Engineering *Spring 2026 (∼38 students)*

Applied ordinary differential equations for biomedical engineering students, covering integrating factors, systems of ODEs, and second-order constant coefficient equations.

MATH 243: Analytic Geometry and Calculus C *Fall 2025 (∼80 students)*
MATH 242: Analytic Geometry and Calculus B *Summer 2025 (∼30 students)*
MATH 243: Analytic Geometry and Calculus C *Summer 2024 (∼30 students)*
MATH 242: Analytic Geometry and Calculus B *Winter 2024, two sections (∼30 each)*

Responsibilities included delivering lectures, designing assignments and exams, and assessing student performance.

Teaching Assistant — Department of Mathematical Sciences, University of Delaware

MATH 637: Mathematical Techniques in Data Science *Fall 2024, Spring 2024*
MATH 219: Data Science I *Spring 2024*
MATH 243: Analytic Geometry and Calculus C *Fall 2023*
MATH 242: Analytic Geometry and Calculus B *Spring 2023*
MATH 241: Analytic Geometry and Calculus A *Fall 2022*

Responsibilities included conducting discussion sessions, grading coursework, and holding office hours.

Technical Skills

Programming Languages: Python (Proficient), Julia (Intermediate)

Machine Learning and Scientific Computing: PyTorch, TensorFlow, Keras, DeepXDE, BayesFlow, Scikit-learn, NumPy, SciPy

Scientific Machine Learning and Numerical Methods: Neural Operators (DeepONet, Fourier Neural Operators), Diffusion Models, Physics-Informed Neural Networks (PINNs), Neural ODEs, Transformers, Fourier Neural Networks, Finite Element Methods, Optimization, Numerical Solution of Differential Equations

Computing and Development Tools: Git, Linux, DARWIN HPC Cluster, VS Code, Anaconda, Jupyter Notebook
Data Formats and Workflow Tools: HDF5, YAML, SLURM

Honors and Awards

Institution Silver Medal — IISER Berhampur *2021*
Awarded for highest cumulative grade point index in the Department of Mathematical Sciences.

INSPIRE Fellowship — Government of India *2016 – 2021*
National merit-based fellowship awarded to the top 1% of students for pursuing higher education in science.

Conferences and Workshops

SIAM Annual Meeting (AN26), Cleveland, OH *July 2026*
Presentation: *Can a single neural network generate samples from an entire family of attractors — from steady to fully chaotic?*

Mid-Atlantic Numerical Analysis Day, Temple University *November 2025*
Presentation: *Operator Learning for Tear Film Dynamics*

Graduate Student Mathematical Modeling Camp (GSMMC), *June 2023*
University of Delaware

Mathematical Problems in Industry (MPI), New Jersey Institute of *June 2023*
Technology

Languages

Bengali — Native | English — Fluent | Hindi — Conversational

References

References available upon request.